

Task 19: Next Greater Element I

Problem Description

The next greater element of some element x in an array is the first greater element that is to the right of x in the same array. You are given two distinct 0-indexed integer arrays `nums1` and `nums2`, where `nums1` is a subset of `nums2`. Return an array `ans` of length `nums1.length` such that `ans[i]` is the next greater element as described above.

Java Solution

```
class Solution {
    public int[] nextGreaterElement(int[] nums1, int[] nums2) {
        Map<Integer, Integer> map = new HashMap<>();
        Stack<Integer> stack = new Stack<>();

        for (int num : nums2) {
            while (!stack.isEmpty() && stack.peek() < num) {
                map.put(stack.pop(), num);
            }
            stack.push(num);
        }

        int[] ans = new int[nums1.length];
        for (int i = 0; i < nums1.length; i++) {
            ans[i] = map.getOrDefault(nums1[i], -1);
        }
        return ans;
    }
}
```